Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)

Munshi Nagar, Andheri (W), Mumbai – 400 058.

End Semester Examination
Jul 2025

Max. Marks: 100 **Duration: 180 Min.**
Class: Final Year BTech(CE) **Semester: VII**
Course Code: CS401 **Program: BTech(CE)**
Name of the Course: Cryptography and System Security

Instruction:
1) **All questions are compulsory.**
2) **Draw neat diagrams and write equations wherever necessary.**
3) **Assume suitable data if necessary but justify the same.**

Course Outcomes: At the end of the course, student will be able to:
(1) Describe the different types of the cryptographic algorithms to secure information.
(2) Apply different cryptographic techniques to solve security-related problems
(3) Create a message digest from data to authenticate authorized user
(4) Use system security practices

Q. No.	Question	Max. Marks	CO-BL
Q. 1A	This problem explores the use of a one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5 ..., then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on. (i) Encrypt the plaintext *sendmoremoney* with the key stream 9 0 1 7 23 15 21 14 11 11 2 8 9. (ii) Using the ciphertext produced in part (i), find a key so that the cipher text decrypts to the plaintext *cashnotneeded.*	**10**	CO1-BL3
Q. 1B	List five types of cryptanalytic attacks based on what is known to the attacker and briefly discuss all listed types.	**10**	CO1-BL2
Q. 2A	In an RSA system, the public key of a given user is e = 31, n = 3599. What is the private key of this user? First use trial-and-error to determine p and q; then use the multiplicative inverse of 31 modulo f(n) to get final results. Show small examples for encryption and decryption.	**10**	CO1-BL3
Q. 2B	Consider a Diffie-Hellman scheme with a common prime q = 11 and a primitive root $\alpha = 2$. (i) Show that 2 is a primitive root of 11. (ii) If user A has public key $Y_A = 9$, what is A's private key X_A? (iii) If user B has public key $Y_B = 3$, what is the secret key *K* shared with A?	**10**	CO2-BL4
Q. 3A	What four requirements were defined for Kerberos? In the context of Kerberos, what is a realm?	**10**	CO2,3-BL2
Q. 3B	Is it necessary to recover the secret key in order to attack a MAC algorithm? Justify your answer formally with an example or a particular situation. **OR** What is the difference between weak and strong collision resistance? Discuss weak and strong collision resistance properties of hash function in detail with a small example.	**10**	CO4-BL3

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Q. 4A	Answer the following questions about tag values in PGP: (i) Can a packet with a tag value of 1 contain another packet? (ii) Can a packet with a tag value of 6 contain another packet? Justify your answer by discussing the packet details corresponding to (i) and (ii).	**10**	CO2,3-BL4
Q. 4B	What are the differences among the three configurations of firewall? Discuss all the details of three firewall configurations through architecture/design-level/block-level diagram.	**10**	CO4-BL2
Q. 5A	Define AH and ESP. Discuss the security services provided by AH and ESP.	**10**	CO4-BL2
Q. 5B	When a session is resumed with a new connection, does SSL require the full handshaking process? Justify your answer by showing the messages that need to be exchanged in a handshaking.	**10**	CO4-BL4